

KUNJ RATHOD

kunjrathod.com | +1 (385) 202-8879 | kunj.rathod@utah.edu | linkedin.com/in/rathodkunj | github.com/rathodkunj2005

Education

University of Utah

B.S. in Computer Science GPA: 3.7/4.0 — Dean's List

Aug 2023 – Dec 2026

Salt Lake City, UT

– Coursework: Computer Vision, NLP, Machine Learning, Distributed Systems, Operating Systems, Algorithms

Research

Spatial Memory for Embodied Agents & Long-Horizon Agents

Jan 2026 – Present

University of Utah

- Designing retrieval-augmented spatial memory architectures for embodied agents, unifying spatial, temporal, episodic, and semantic memory for long-horizon robot navigation and manipulation.
- Automatic simulation environment generation from natural language (SceneSmith, MuJoCo) for scalable zero-shot robot policy evaluation; benchmarking long-horizon agents (OS Marathon, BrowseComp, Mind2Web, LongMemEval).

Evaluation Infrastructure for Long-Context Retrieval (MEMROT)

Jan 2026 – Present

University of Utah – CHPC

- Built distributed evaluation pipeline for mechanistic analysis of long-context retrieval failure in Gemma-2-2B: 840-configuration activation-patching sweep orchestrated via SLURM on university HPC; stood up vLLM serving on shared GPU clusters for large-scale agent benchmark runs.

Circuit-Level Generalization in LLMs via Sparse Feature Circuits

Jan 2026 – Present

University of Utah

- Automated pipelines extracting sparse feature circuits from default tasks to predict generalization to non-default tasks via attribution graphs; analyzing circuit stability across model families.

Multi-Agent Materials Discovery for Extreme Environments

Aug 2025 – Feb 2026

STARS Lab, University of Utah (Collaboration: NASA, Microsoft, U.S. DoD)

- Multi-agent, graph-augmented pipeline normalizing material-property data from 1,000+ papers into a physics-aware knowledge graph; constraint-based design engine enforcing temperature, creep, and pressure limits for aerospace material selection.

Industry Experience

Software Engineering Intern — Fabric Enterprise Platform, Azure Data

May 2026 – Present

Microsoft

Redmond, WA

- Shipped SKU Change History feature end-to-end on Microsoft Fabric: C# distributed backend services, React/TypeScript frontend, and MDM telemetry integration; delivered behind feature-flag rollouts across enterprise capacity-management infrastructure.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health

Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference with p95 <200ms TTFT, 40% latency reduction via pipeline optimization and API caching; distributed session persistence (DynamoDB + S3) for 1,000+ concurrent conversations.
- Shipped 6 features across 4 sprints (Bedrock Agents, Knowledge Bases, Guardrails); owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS).

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen

Remote

- Scaled hybrid retrieval (dense + BM25 + reranking) to 10M+ documents (+28% recall, –35% hallucinations); benchmarked 8 LLM families on F1/latency/cost across LegalBench and NyayaAnumana; model routing cut inference spend.

Projects

FNDR — On-Device Multimodal Inference for macOS

Rust, Tauri, Metal, Apple Vision

- Built fully on-device semantic file search app: Metal-accelerated local model inference and Apple Vision Framework OCR, no cloud dependency; optimized for latency and memory on consumer Apple Silicon hardware.

Continuum — Local-First Agent Memory Engine & Developer Tooling

Rust, Tauri, LanceDB, MCP

- Designed local-first long-term memory engine for AI agents with embedded vector store (LanceDB) and an MCP server exposing memory APIs to any agent client, making the system usable by external developers and tools.

HirePilot — Autonomous AI Recruiting Pipeline (Hackathon Winner)

Python, Claude API, Twilio

- 9-agent agentic pipeline: Slack-triggered GitHub sourcing, enriched candidate profiles, live adaptive voice interviews via Twilio + Claude, multi-dimensional scoring with ranked shortlist delivery; shipped in 24 hours.

FlowVia — V2X Edge Traffic Optimization

Python, TensorFlow (LSTM), DSRC/C-V2X, OBD-II

- Real-time speed recommendation from live SPaT sensor data; LSTM traffic prediction on streaming data; AES-256 encrypted V2V/V2I communication with edge-first processing for sub-10ms latency.

Technical Skills

Languages: Python, C++, Rust, Java, TypeScript, C#, SQL

AI/ML: PyTorch, vLLM, Sparse Feature Circuits/SAEs, RAG/GraphRAG, Multi-Agent & Agentic Systems, Embodied AI

Systems & Infra: SLURM/HPC, Distributed Evaluation Pipelines, AWS (Lambda, Bedrock, S3, DynamoDB, RDS), Docker, Kubernetes, gRPC, CI/CD, On-Device Inference (Metal)

Robotics & Simulation: MuJoCo, Scene Generation (SceneSmith), Spatial Memory, Real-Time Sensor Fusion, V2X